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Artificial intelligence in sport: A narrative review of applications, challenges and future trends

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ABSTRACT

This narrative review explores the transformative impact of artificial intelligence (AI) in sport, covering its applications, challenges and future directions across key areas such as biomechanics, performance enhancement, sports medicine, health monitoring, coaching and talent identification. AI can potentially empower athletes to optimise movement, personalise training, improve diagnostics and accelerate rehabilitation. However, integrating AI into sport presents challenges, particularly around data privacy, ethical concerns and adoption within sport organisations. This review also addresses these issues, highlighting strategies for responsible data governance and transparency. Furthermore, the review explores the promising future trends for AI in sport, which suggest a profound impact how sport is practiced and managed globally, pointing towards an era of enhanced performance, health and inclusivity.

KEYWORDS

Machine learning; biomechanics; performance enhancement; sports medicaine and health; talent identification; data and ethical considerations

Introduction

Sport has evolved into a data-rich field where technology integrates seamlessly with athletic performance, health and strategy (Hutchins, 2016). This transformation is firmly grounded in the long-standing tradition of using data and evidence to enhance sporting outcomes (Vincent et al., 2009). From the early days of manually recorded performance metrics to the adoption of wearable sensors (Yang, 2024) and video analysis (Rangasamy et al., 2020), the sporting world has consistently embraced tools that provide actionable insights.

As the volume and complexity of available data continue to expand, traditional methods of analysis are reaching their limits (Donoho, 2000). The opportunity to process vast datasets, uncover subtle patterns and make real-time decisions has paved the way for the integration of advanced technologies (Malekloo et al., 2022). Artificial intelligence (AI), with its ability to learn from data, recognise complex patterns and predict outcomes (Jiang et al., 2017), represents the natural progression of this journey. By building on the foundations of data, evidence and analytics, AI is opening new opportunities to athlete performance, training optimisation, injury prevention and personalised coaching (Guelmami et al., 2023; Pickering & Kiely, 2019).

In this review, AI in sport refers to the application of advanced computational systems and algorithms designed to process, analyse and act upon vast and complex datasets, extending beyond algorithms to include the integration of sensors, which capture real-time physiological and biomechanical data, and effectors, which translate AI-driven insights into physical responses or actions. Machine learning algorithms, for instance, play a central role in predicting injury risks (Owen et al., 2024), modelling performance metrics (Bunker & Susnjak, 2022; Cust et al., 2019) and identifying talent (Musa et al., 2020; Reyaz et al., 2022), enabling data-driven decisions for both athletes and coaches. Deep learning models are particularly valuable in sports medicine (H. Ma & Pang, 2019; Ramkumar et al., 2022), assisting in diagnostics (Wu et al., 2022) and supporting personalised treatment planning (Parker & Forster, 2019). Ensemble learning, by integrating multiple models, increases diagnostic accuracy and strengthens recommendations in health monitoring (Ali et al., 2020). Natural language processing (NLP) – a field of AI focused on enabling machines to understand and respond to human language – further aids coaching by providing real-time feedback through AI-based chatbots and improving athlete-coach communication

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This article has been corrected with minor changes. These changes do not impact the academic content of the article.

(Boughattas et al., 2022). Computer vision techniques offer insights into athletes' biomechanics (Colyer et al., 2018) by capturing and analysing movements, allowing detailed assessments of technique and injury risks. Predictive analytics informs strategic decision-making (Watanabe et al., 2021), generating real-time insights into player performance and game dynamics. Augmented reality (AR) (Lee et al., 2023) and virtual reality (VR) (Witte et al., 2022) create immersive environments for training and rehabilitation, aiding skill improvement and effective recovery (Patil et al., 2022). Together, these components form a sophisticated ecosystem where AI is not only a computational tool but also a critical enabler of precision and personalisation in sport.

Despite the potential benefits of AI in sport, significant challenges and gaps remain. Key issues include ethical concerns (Trail et al., 2024) around data privacy and potential biases in AI algorithms, which might lead to discrimination or unfair advantages (Najjar, 2023, Yan, 2022). Integration challenges also arise as sports organisations grapple with the costs and complexities of adopting new AI technologies, disrupting traditional methods (Zhang et al., 2023). Furthermore, the variability in AI performance in real-world sports settings raises questions about the reliability and robustness of these technologies (Araújo et al., 2021).

This narrative review explores the applications of AI in sports across the key domains: biomechanics, performance enhancement, sports medicine, health monitoring, and coaching and talent identification. These areas represent some of the most impactful and promising uses of AI, with significant potential to advance sports science and practice. Closely aligned with current research trends and industry priorities, they are highly relevant for both academic exploration and practical application. Key challenges, such as ethical and privacy concerns, data governance and the complexities of real-time AI applications, are also addressed. Finally, the review discusses promising future trends for AI in sports, emphasising the importance of responsible integration to maximise benefits while mitigating potential risks.

AI applications in sport

The review highlights AI's role in biomechanics, focusing on movement optimisation, technique refinement and injury prevention, as well as its contributions to performance enhancement through advanced training protocols, strategy development and performance analytics. AI's impact on sports medicine is discussed, particularly in diagnostics, personalised treatment and rehabilitation, along with its applications in health monitoring, including nutrition planning, mental well-being and athlete wellness. Additionally, it explores how AI-driven insights are transforming coaching strategies and talent identification.

AI in biomechanics: Enhancing movement and reducing injuries

Biomechanics focuses on describing motion (kinematics) and the factors driving it (kinetics) (Hamill et al., 2021). Analysing biomechanical data supports improvements in movement patterns, technical skill execution and injury reduction (Hamill et al., 2021). Technology has played a crucial role in sports biomechanics especially when examining highly dynamic motions that occur over short time-periods such as foot contact during sprinting or ball release in striking and throwing activities, which all require high-frequency data collection (sometimes exceeding 1000 samples per second). Without such high-frequency data collection, there will be insufficient time-points to accurately describe the movement's displacement, which further magnifies errors when calculating other kinematic and kinetic variables. Therefore, even with technological advances, standard 3D motion capture and analysis remain complex and time-consuming (Hamill et al., 2021)

AI is increasingly applied in biomechanics to improve (1) data collection methods outside laboratory settings and (2) the ability to integrate different types of data to make more accurate predictions about important outcomes including injury risk (Bennett et al., 2019; Cornish et al., 2024; Giarmatzis et al., 2020; Haller et al., 2023; Lövdal et al., 2021; Sharma et al., 2023; Zhu et al., 2021). One notable application is the use of AI to evaluate gym-based exercises, such as the squat (Hu et al., 2024). Inertial measurement units (IMUs) and a gated long short-term memory transformer AI network were used to classify correct and incorrect squat techniques among 22 male participants, achieving a notable 96.3% accuracy in detecting proper form (Hu et al., 2024). Such AI methods, if integrated into mobile apps, could improve exercise techniques and reduce injury risk for fitness enthusiasts (Chariar et al., 2023; Zheng et al., 2023).

AI has also been applied to estimate injury risk at the knee and hip during common movements such as walking and running (Benjaminse et al., 2024; Cornish et al., 2024; Sharma et al., 2023; Van Hooren et al., 2024). An artificial neural network (ANN) model was used to estimate tissue loads at common injury sites, achieving error rates for patellofemoral stress, tibial stress and Achilles tendon strain impulses of $1.95 \pm 8.40\%$, $-7.37 \pm 6.41\%$ and $-12.8 \pm 9.44\%$, respectively (Van Hooren et al., 2024). The study, which included treadmill running with varied step frequency and trunk lean, used ARION pressure insoles and 3D motion capture for input variables.

Computer vision plays a crucial role in athlete training and assessment by enabling non-intrusive, real-time analysis of movement patterns, technique and biomechanics. Using video footage from cameras or drones, AI algorithms can automatically track body posture, joint angles and movement sequences to evaluate performance and detect deviations linked to injury risks (Barris & Button, 2008). This technology is increasingly integrated into training environments to provide coaches with precise feedback without relying on manual observation or costly motion capture systems.

While many studies show AI's potential for accurate predictions, the required data inputs, such as force plate data, IMUs and 3D motion capture, limit accessibility beyond elite sports settings. This highlights the need for simpler, affordable AI-based solutions, like smartphone video analysis, to make these insights available to recreational athletes (Amrein et al., 2023; de Vries et al., 2022).

Future research should focus on validating AI's use in dynamic tasks, such as sprinting and jumping, and enhancing model accuracy by incorporating individual-specific data (Stetter et al., 2020). Commercial collaborations, like those with ARION, which offers validated AI-based insoles for runners, may also advance the practical application of AI in biomechanics. Simplified, accessible tools could broaden the impact of AI in sport, helping athletes at all levels prevent injuries and optimise movement.

Pushing boundaries: AI-powered performance enhancement

Optimal performance in sports results from years of specialised training, dietary and recovery processes, and strategic implementation (Kellmann et al., 2018). Collecting, analysing and interpreting training and competition data, now enhanced by AI, has the potential to further improve performance (Forcher et al., 2023; García-Aliaga et al., 2023). Performance analysis is particularly critical in team invasion sports, which are recognised as dynamic systems where opposing teams interact (Lord et al., 2020). Six primary themes in team sport behaviour have been identified as focal areas in performance analytics: (i) game actions, (ii) dynamic actions, (iii) movement patterns, (iv) collective team behaviour, (v) social network analysis and (vi) game styles (Lord et al., 2020). These themes could be positively influenced by AI through (i) automated data analysis, (ii) uncovering complex patterns and (iii) providing real-time insights to enhance decision-making and performance strategies (Lord et al., 2020).

Traditionally, sport science research examined specific training aspects, such as improving physical capacities like strength, sprinting and aerobic power (Huiberts et al., 2024; Swinton et al., 2024). Research is increasingly focusing on match performance data, including GPS-based movement characteristics (Griffin et al., 2021; Janetzki et al., 2021; Supej et al., 2019), skill execution (Bruce et al., 2018; Robertson et al., 2016; Yue et al., 2014) and tactical behaviour (Browne et al., 2021; Lord et al., 2020; Plakias et al., 2023). Despite advances, processing this data remains time-intensive, often delaying real-time feedback that could provide quantitative data to support coaches' tactical decision-making during competitions (Lord et al., 2020; Pradas de la Fuente et al., 2023; Zhang & Leng, 2024). Consequently, using AI to provide real-time performance feedback in sport and rehabilitation contexts has much application, with examples including spinal cord injury rehabilitation (Tao et al., 2024) and for measuring barbell velocity in the bench press (Balsalobre-Fernández et al., 2023).

AI offers a solution to these challenges by optimising training protocols (Washif et al., 2024) and monitoring training components, especially in resistance training (Chariar et al., 2023; Hu et al., 2024; Zhu et al., 2021). Its strength lies in efficiently analysing large datasets encompassing physical location and movement (Born et al., 2024; Marquina et al., 2023), technical skills (García-Aliaga et al., 2023; Song et al., 2023) and tactical behaviour (Forcher et al., 2023; García-Aliaga et al., 2023), thereby addressing the limitations of traditional data processing methods.

For instance, Born et al. (2024) tested the You Only Look Once (YOLO) AI model's ability to identify opposing players' locations in Australian Rules Football, using data from 17 matches in the 2019 AFL season. The model achieved high accuracy (mAP = 0.94; precision = 0.95; recall = 0.97; F1-score = 0.96), with minimal error in player position estimates compared to GPS data, indicating YOLO's utility for tracking opponents and enhancing tactical preparation.

Forcher et al. (2023) used player location data from the 2020–2021 German Bundesliga season to predict defensive success in football (i.e., regaining possession). Among the classifiers tested, a random forest model with 16 features performed best (accuracy = 0.82, precision = 0.47, recall = 0.70, F1-score = 0.57). This model identified critical defensive tactics like ball pressing and compact organisation near the ball. By mapping performance metrics to contextual factors such as ball location, AI can deepen insights into team behaviours, supporting advanced performance analysis (Browne et al., 2021; Lord et al., 2020).

Future research could explore digital twin technology to model opponents and devise optimal counter-strategies. A digital twin (Singh et al., 2021) – a virtual representation of an opposing team – could simulate performance and tactical behaviour using historical and real-time data, helping coaches identify the best team compositions and strategies.

AI could also suggest tailored tactics by analysing opponents' patterns, such as movement trends, skills and tactics (Wang et al., 2024). These advancements could enhance decision-making, adaptability and competitiveness, both in preparation and during matches (Chen, 2024).

Revolutionising sport medicine: AI in diagnosis, treatment and rehabilitation

AI has shown rapid progress in numerous disciplines, with sports, medicine and health care being no exception (Fares et al., 2024; Ramkumar et al., 2022; Sulaiman, 2024). AI brings a promising shift in sports medicine by providing opportunities for injury prevention, injury recovery planning and rehabilitation techniques (Ramkumar et al., 2022). Sports medicine, mainly concerned with the diagnosis, treatment and rehabilitation of athletes, has evolved into broad fields comprising diagnostic decision support system (DDSS) (Rigamonti et al., 2021), health apps (Dergaa et al., 2024), smart trackers/devices (Chidambaram et al., 2022; Yang, 2024), telemedicine (Lal et al., 2023), augmented reality (Wang et al., 2025), virtual reality and exergaming (Donath et al., 2016; Lal et al., 2023). Athletes suffer from mild-to-severe injuries and their timely recovery can be beneficial in terms of cost and performance (Rigamonti et al., 2021). Sports medicine provides necessary medical facilities to the athletes, and AI can improve these processes through timely and insightful decisions (Pareek et al., 2024).

AI is increasingly being adopted to address numerous challenges in sports medicine (Ramkumar et al., 2022b). Traditional methods of sports medicine for athletes rely on teams of specialised doctors, orthopaedists and physiotherapists and can be limited by personal bias, high costs, inconsistent availability and difficulty in tailoring care for rare or complex cases (Ramkumar et al., 2022b; Reis et al., 2024; Smaranda et al., 2024). AI has the potential to resolve these issues by providing numerous applications in sports medicine (Fayed et al., 2023; Reis et al., 2024), including (i) injury prevention, (ii) rapid diagnosis, (iii) rehabilitation and recovery (Guelmami et al., 2023) and (iv) real-time personalised treatment planning (Pavunraj et al., 2025). AI systems rely on substantial amounts of data to improve their generalisability (Kelly et al., 2019); with big data from fitness, exercise and sports medicine devices helping AI models provide personalised diagnosis and treatment plans (Dilsizian & Siegel, 2014). But these systems face challenges like data privacy, data noise, domain shift over time, smart device quality and model capacity (Chidambaram et al., 2022; Rigamonti et al., 2021; Zeng et al., 2021).

The vast amount of data generated by smart devices can be too complex for physicians to interpret quickly and use effectively in accurate diagnosis and rehabilitation planning (Pareek et al., 2024). AI can help understand hidden trends faster, providing a second opinion to the physician and helping focus on the core issue faced by the athletes (Pareek et al., 2024; Qazi & Iqbal, 2024; Sulaiman, 2024). For example, 'VICTOR' is a recently developed custom GPT-based model, specifically tuned for sports medicine problems (Naughton et al., 2024; Valencia, 2024). Another example is that of ACL rupture model which significantly improved the diagnostic performance of sports medicine experts from 96% to 98% and sport medicine trainees from 84% to 96%, validating the significance of second opinion of AI models (Wang et al., 2024). AI systems can help physicians devise personalised therapy and treatment plans by focusing on apparent as well as hidden factors such as medical history, injury type, location, severity, real-time treatment response and multi-modal data from smart devices (Chidambaram et al., 2022; Qazi & Iqbal, 2024). AI-based personalised treatment plans, providing accurate and comprehensive recovery strategies, can improve the chances of a successful recovery of an athlete and can cut down time and financial burden on the team (Qazi & Iqbal, 2024; Zhan, 2024).

Rehabilitation can also assist athletes recovering from spinal cord and brain-related injuries (X. Wang et al., 2025; Yang et al., 2022). Athletes can suffer from severe head and spinal pathologies (Ramkumar et al., 2022). AI is revolutionising rehabilitation by simultaneously analysing complex interactions between physiological markers, rehabilitation history and an athlete's current performance (Cui, 2024); thus tailoring rehabilitation programmes accordingly (Ramkumar et al., 2022). A recent comparative study showed the significance of RNN-LSTM model in predicting rehabilitation outcomes for sports injuries (Cui, 2024). AI algorithms providing real-time feedback and personalised treatments expedite the healing process and minimise the risk of future injury, enabling athletes to return to competition stronger and more resilient (Guelmami et al., 2023; Zeng et al., 2021).

Recently, there has been a growing trend in utilising chatbots and diagnostic decision support systems (DDSSs) to assist in diagnosing minor and common medical issues (Dergaa et al., 2023; Fayed et al., 2023; Qazi & Iqbal, 2024; Rigamonti et al., 2021). These systems estimate the severity of the injury and help players as well as their trainers/physicians to take appropriate actions in due time (Rigamonti et al., 2020). Such tools are particularly useful in cases when the athlete's trainer and physician is not readily available on-site or busy with other severely injured players (Rigamonti et al., 2021). These chatbots and DDSSs, equipped with expert knowledge, can also be used by medical specialists in cases where they might be unsure about the state of the player or what actions to take (Fayed et al., 2023; Rigamonti et al., 2021).

Para athletes can particularly benefit from AI-driven innovations in sports medicine, as their rehabilitation and treatment often require highly individualised and adaptive approaches (Rosa, 2025; Rum et al., 2021). AI-powered prosthetics and orthotics can learn from the user's movement patterns and adapt in real time to improve functionality, comfort and performance (Dyer et al., 2010). These adaptive technologies not only support better medical outcomes but also enhance autonomy and long-term athlete development in Para sport (Teixeira & Alves, 2021). Ensuring these AI systems are inclusive and sensitive to classification-based differences remains a key research and ethical consideration.

Integrating AI into sports medicine offers significant potential, yet AI-driven systems currently lack the nuanced reasoning, intuition and contextual understanding essential for complex clinical decision-making (Naughton et al., 2024a; Reis et al., 2024; Rigamonti et al., 2020). Furthermore, many AI algorithms are 'black box' in design, which restricts openness and trust and makes clinicians reluctant to rely on results they are unable to completely understand or defend (Mennella et al., 2024, Najjar, 2023). Another significant issue is accountability; it's not apparent who is at fault if an AI-informed choice causes harm – the doctor, the developer or the organisation (Mennella et al., 2024; Sulaiman, 2024). Despite these limitations, AI's strengths in pattern recognition, data processing and diagnostic support position it as a valuable collaborative tool. AI can improve clinical workflows when utilised as a second opinion system; studies show that AI-human collaborations frequently perform better than either one alone (Wang et al., 2024; Cui, 2024).

AI offers significant potential to uplift the sports medicine industry. Its ability to discover and learn existing patterns and generalise to unseen patterns can provide significant insights to sports medicine practitioners and complement their judgements. The future direction of AI in sports medicine is to provide explainable and reliable diagnosis results (XAI), specialised treatment plans and real-time rehabilitation support (Mennella et al., 2024; Pareek et al., 2024). It is also important to address ethics and practical factors to make AI systems more reliable (Mennella et al., 2024). AI-based tools for sports medicine are highlighted in Figure 1.

Health monitoring with AI: Supporting athlete wellness and nutrition

Physical and mental health are very important for everybody but are especially important for elite athletes whose overall health and well-being directly impact their sporting performance and financial livelihood (O'Donnell et al., 2024; Puce et al., 2024). Physical health is often associated with an individual's outward appearance and physical capabilities (Berry, 2016), while mental health pertains to the more abstract psychological processes that influence a person's well-being (Taylor & Brown, 1988. Both are crucial for the success and longevity of an athlete's career (Reardon et al., 2021). The overall wellness of an athlete relies on balancing their training, competition, nutrition and recovery strategies (Mirmomeni et al., 2021; Reardon et al., 2021).

Recently, large language models (LLMs) like ChatGPT and Gemini have made a significant impact on various industries including sports (Agne & Gedrich, 2024; Papastratis et al., 2024). These LLMs have demonstrated performance or capabilities comparable to nutritionists, doctors and sports health practitioners, such as (i) providing advice (Agne & Gedrich, 2024), (ii) analysing data (Shool et al., 2025) or (iii) generating insights (Qiu et al., 2024). One such study shows that ChatGPT performed better than a well-established tool in the market (Agne & Gedrich, 2024). Another study created ChatGPT-based system to provide precision nutrition recommendations (Papastratis et al., 2024). A health monitoring system, WHMSHAR, provides an end-to-end system for the wellness of athletes (Y. Yang, 2024b). It uses wearable technology with numerous sensors to collect athlete's physiological data then uses an AI algorithm to examine multi-modal data for and provide rapid feedback and tailored suggestions (Y. Yang, 2024b). The study shows that combining wearable sensor technologies with AI presents a viable strategy for thorough health monitoring in athletic environments (Y. Yang, 2024b). Similarly,

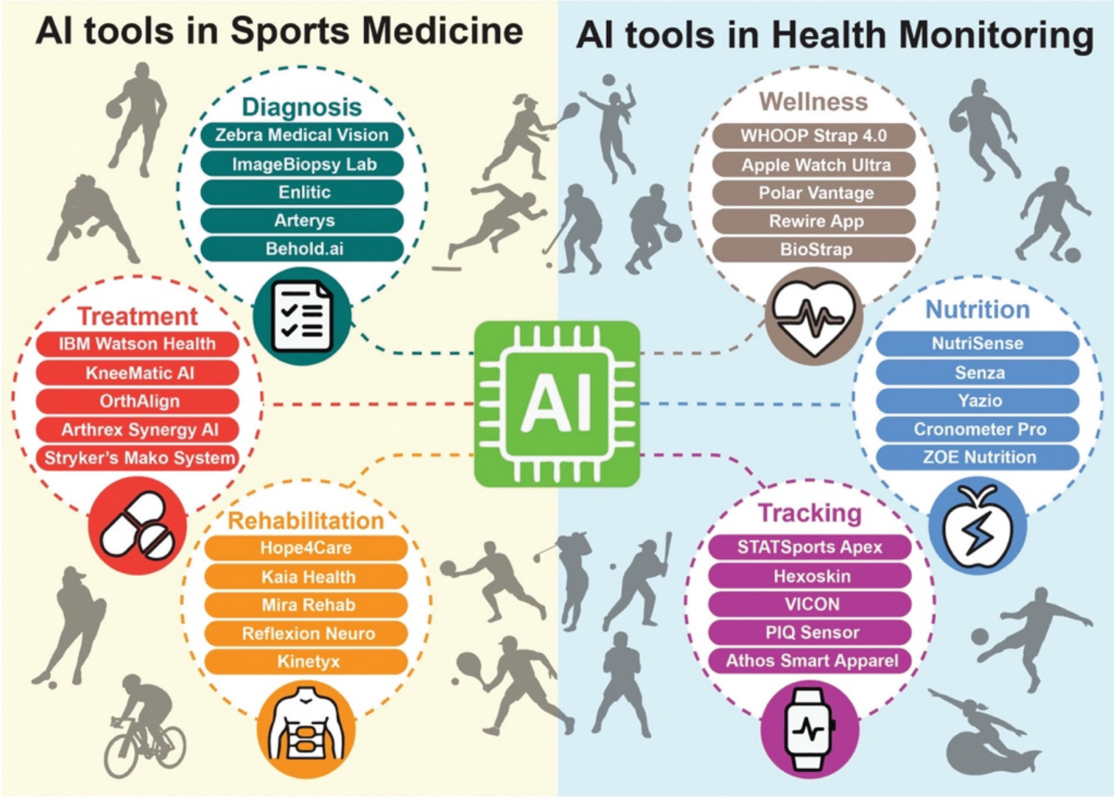

Figure 1. AI based tools for sports medicine (left) and health monitoring (right). These tools can be used by athletes and coaches to help them in their daily workflow.

Intelligent Garment Systems are embedded with sensors that capture physiological data such as heart rate, muscle activity and other bioelectric indicators to track an athlete's physical and emotional states (Shen et al., 2023). An on-chip AI models process these continuous signals and provide real-time feedback and suggestions to avoid injuries and adverse health conditions (Shen et al., 2023).

AI can help health practitioners by analysing data quickly (Liang et al., 2022; Tong & Ye, 2023; Zhen et al., 2021). Future research and applications should aim to combine the insights generated by AI systems with expertise of health practitioners to develop personalised plans for nutrition, physical activity and mental well-being (Lu, 2024; Mirmomeni et al., 2021; Reardon et al., 2021). LLMs are very sensitive to context and prompts, with many current health monitoring studies based on a single prompt (Dergaa et al., 2023). This limited approach makes it difficult to assess how LLMs perform across varying user interactions or compare them meaningfully to expert-driven or traditional health monitoring methods. Therefore, new research should focus on iterative and multi-prompt engagements with LLMs to simulate more realistic and complex health scenarios. Such comparative evaluations can reveal where LLMs align with or diverge from expert recommendations, identify knowledge gaps and help assess their reliability, ultimately informing both technical improvements and ethical standards. Some tools that provide AI-based health monitoring service are highlighted in Figure 1.

Next-level coaching and talent identification with AI insights

AI is increasingly being integrated into coaching techniques (Jud & Thalmann, 2025) and talent identification (McAuley et al., 2024) in sports, revolutionising traditional approaches with data-driven insights (Chmait & Westerbeek, 2021), improved performance analysis (Araújo et al., 2021) and enhanced decision-making capabilities (Janssen et al., 2023).

AI's role in coaching focuses on providing deeper insights into athletes' performance, developing personalised training programmes and enabling real-time feedback. Some key AI techniques in coaching include performance analysis (Araújo et al., 2021; Yang & Chang, 2023), predictive analytics (Sharma et al., 2022), virtual reality (Witte et al., 2022) and augmented reality (Lee et al., 2023), and natural language processing (NLP) for feedback (Boughattas et al., 2022).

Talent identification is a crucial part of athlete development (Höner et al., 2023), aiming to spot young athletes with the potential to excel in their respective sports. Traditionally, it has been driven by subjective assessments from coaches and scouts, often limited by human biases and regional constraints (Lawlor et al., 2021). AI offers new ways to automate, analyse and enhance the talent identification process, bringing data-driven objectivity (Thirunagalingam et al., 2025). With capabilities like machine learning, computer vision and predictive analytics, AI can assess athletes in ways that were previously impractical (Ghosh et al., 2023). For example, platforms like AiSCOUT (ai.io, 2025) use computer vision to remotely evaluate athletes' movement patterns and skills, while the Australian Institute of Sport (AIS) has explored AI models to identify talent based on junior athlete performance data (Australian Institute of Sport, 2025). Machine learning algorithms can analyse data such as physical metrics (Teunissen et al., 2023), performance statistics (Jamil et al., 2021) and historical match records to identify patterns indicative of high potential (Bunker & Susnjak, 2022). AI systems can also process video footage (Bennett et al., 2019) to analyse an athlete's technical skills, such as ball control in soccer, shooting mechanics in basketball or stroke technique in swimming. For example, NBA clubs leverage the AI systems implemented by Kinexon to monitor and assess athlete performance (Gálvez et al., 2024), using wearable sensors (Elis, 2024) to track metrics such as player load, acceleration, speed, distance covered, heart rate and positional data. This highlights the growing integration of AI in professional sports.

The effectiveness of AI models is closely tied to the quality of input data, and poor or inconsistent datasets can significantly undermine their reliability (Chmait & Westerbeek, 2021). In addition, the collection of detailed biometric and performance data raises important ethical and privacy concerns – these are explored in more depth in the Challenges and Recommendations for Trustworthy AI in Sport section. Cost is another major barrier to adoption, particularly for smaller sports organisations. Advanced AI-driven video analysis systems, for instance, can incur annual costs of tens of thousands of pounds in licensing and infrastructure, whereas more traditional methods like manual video review or stopwatch timing involve minimal equipment and staffing costs (Naughton et al., 2024).

AI in coaching and talent identification offers significant benefits, including objectivity and consistency. By reducing human bias, AI enables more accurate and impartial evaluations of talent (Lee & Lee, 2021). Moreover, AI has the potential to democratise talent identification by reaching underserved and remote regions around the globe, where traditional scouting methods may be limited. Companies like PlayerMaker (https://www.playermaker.com) and ai.io (https://www.ai.io) are already leveraging AI-driven tools to assess and identify talent in hard-to-reach areas, providing young athletes with opportunities to showcase their skills and access tailored development programmes. By enabling earlier interventions and customised training, AI can help uncover hidden talent and promote a more inclusive approach to athlete development (Bennett et al., 2019).

However, the adoption of AI in coaching and talent identification is not without challenges. Coaches and scouts may find it challenging to trust AI recommendations if they cannot understand the reasoning behind the predictions (Janssens et al., 2023). This lack of transparency is often due to the complexity of underlying models, such as deep neural networks, which function as 'black boxes' – producing outputs without easily interpretable rationale. Additionally, many AI tools in sport lack built-in explanation mechanisms or user-friendly interfaces that could help practitioners make sense of model decisions. Furthermore, AI models need continuous updates and recalibration to remain accurate, especially as athletes progress and improve over time (Sperlich et al., 2023). The long-term maintenance costs can become a burden if not planned for, impacting the sustainability of AI-based systems.

Challenges and recommendations for trustworthy AI in Sport

This section will explore the critical challenges and recommendations in applying AI within sport, focusing on two key areas: ethical and privacy concerns in handling sensitive athlete data and ensuring data accuracy, consistency and relevance when using wearable technology for real-time analysis. These areas were chosen because they represent foundational issues that influence trust, adoption and the effectiveness of AI in sport. Ethical and privacy concerns are paramount to safeguarding athletes' rights and autonomy, particularly as AI becomes more pervasive in tracking and analysing personal data. Similarly, real-time data accuracy is vital for actionable insights during training and competition, where even minor inaccuracies can have significant implications for performance and decision-making. By addressing these issues through robust governance, enhanced security measures and improved data protocols, the section aims to provide actionable strategies that protect athlete rights while maximising AI's potential in sport performance and decision-making.

Ethical and privacy considerations in AI for Sport

Ethical and privacy concerns are critical when applying AI in sports, particularly regarding sensitive data such as athletes' health metrics, performance statistics and biometric details (Stahl & Wright, 2018). While these datasets are invaluable for improving training and performance, they pose significant risks of unauthorised access, misuse and discrimination (Petersen et al., 2025). For example, biometric data – such as injury risk predictions or physiological stress markers – could be used by teams or sponsors to exclude athletes from competition or contracts, even in the absence of clinical symptoms or athlete consent. Unregulated AI applications could result in intrusive monitoring or data breaches, jeopardising athletes' privacy and autonomy (Najjar, 2023). Recognising these challenges, the Olympic AI Agenda (International Olympic Committee, 2024) highlights the broader ethical and privacy issues associated with AI, including data security, accountability, fairness, job displacement and environmental impact. These concerns underscore the importance of implementing robust frameworks to ensure responsible and transparent AI deployment in sports.

To establish an ethical, privacy-respecting AI environment in sport and build trust, a comprehensive approach is essential. Existing frameworks such as the Organisation for Economic Co-operation and Development (OECD) AI Principles (Organisation for Economic Co-operation and Development, 2024) and the European Commission's Ethics Guidelines for Trustworthy AI (European Commission, 2019) offer foundational guidance on fairness, transparency and accountability, while regulations like the General Data Protection Regulation (GDPR) (European Union, 2016) provide legal standards for data privacy and protection. In the sporting context, frameworks from organisations such as the World Anti-Doping Agency (WADA) also offer sport-specific data governance insights (World Anti-Doping Agency, 2024). Build on these, implementing robust data governance frameworks with clear guidelines on data access and retention supports responsible handling (Orlando, 2022). Advanced encryption and anonymisation further protect data by reducing risks of unauthorised access (Najjar, 2023). Ethical review boards (Guelmami et al., 2023) can evaluate AI projects for biases, while educating athletes on their data rights empowers informed decision-making (Qi et al., 2024).

Several data science methods can also enhance privacy and fairness in sport data. Differential privacy (Abadi et al., 2016; Wasserman & Zhou, 2010), by adding statistical noise, allows models to learn without exposing individual details. Federated learning (T. Li et al., 2020) enables model training on local devices, keeping sensitive data on athletes' personal devices rather than a central server (Zhou et al., 2022). By sending only model updates, not raw data, it enhances privacy. Encryption (Delfs et al., 2002) converts data into an unreadable format, ensuring that only authorised parties with decryption keys can access it. This method safeguards sensitive athlete information both in transit and at rest, preventing potential breaches or leaks (Mcdonald et al., 2016). Anonymisation (Elliot et al., 2018), on the other hand, removes or masks personally identifiable information, making it harder to trace data back to individual athletes. Explainable AI (XAI) (Arrieta et al., 2020) aims to make AI model decisions transparent, showing which factors influence predictions and recommendations. In sport, XAI can clarify how specific metrics, like biometric data or movement patterns, impact injury risk assessments or training advice, helping athletes and coaches make informed, trustworthy decisions (Procopiou & Piki, 2023; Wang et al., 2022). Although these methods show promise, their use in sports data handling is still in its early stages and requires further exploration to fully realise their potential.

Ensuring data accuracy and reliability in real-time AI systems

AI's real-time capabilities are transforming both training and competition in sports, though these advancements present significant challenges related to managing data quality, integration and context-specific interpretation. The growing use of wearable technology exemplifies this shift, as devices like smartwatches, GPS trackers and heart rate monitors feed real-time data into AI systems (Nahavandi et al., 2022). These wearables help track crucial metrics – speed, distance, heart rate and physical strain – throughout training sessions and live events (B. Ma et al., 2020). However, there are challenges in ensuring data accuracy, consistency and relevance across diverse sports contexts. Recommendations include standardising data collection protocols and enhancing sensor reliability to improve data quality and validity, thereby supporting more effective real-time adjustments for load management and injury prevention (Palermi et al., 2024).

In real-time game analysis, AI systems offer insights into tactics, formations and player positioning, allowing for adaptive strategies in sports like football and basketball (Fischer et al., 2019; H. Li et al., 2021). A challenge here is balancing AI-driven insights with the nuanced, situational knowledge of human coaches. Training coaches and sport scientists to interpret and integrate AI insights effectively could enhance in-game decisions without compromising their expertise.

AI's ability to monitor social media in real-time offers significant potential for understanding fan sentiment and driving adaptive marketing and engagement strategies (Midoglu et al., 2024). However, this capability also presents challenges, such as the potential for information overload, which can overwhelm both human teams and AI systems in processing vast amounts of data effectively, and the need to balance insights with privacy considerations. To address these issues, implementing robust filtering tools and establishing clear data-use policies is crucial. These measures not only ensure that teams can focus on actionable insights but also reinforce trust by respecting fans' privacy and preferences. As AI-driven fan engagement becomes an increasingly integral part of sports management, these considerations must align with the broader ethical and governance frameworks applied to AI in sport.

Future trends

Based on the insights and evidence detailed in this review, Table 1 summarises the key broad trends of AI in sport, along with their respective beneficiaries, types of data and the AI methods or tools employed.

Table 1 presents emerging trends in the application of AI in sport, categorised into three key areas: elite sport, recreational sport and physical activity, and research. In elite sport, AI is increasingly used to enhance personalisation, support real-time decision-making and improve performance through technologies such as digital twins and immersive systems. Ethical and data governance considerations are also critical to maintaining trust and compliance in high-performance settings. In recreational sport and physical activity, AI is facilitating broader access to coaching and performance insights through scalable, automated tools, while also enhancing fan engagement through personalised, interactive experiences. Finally, in sport research, AI is advancing areas such as talent identification and the use of simulation for injury prevention and performance modelling, while also prompting important discussions around the ethical use and governance of sensitive data.

Conclusion

AI is poised to deliver important advances across performance, health monitoring, sports medicine, coaching and talent identification in sport. While encouraging developments have already been demonstrated, many applications remain in their early stages and require further validation in real-world contexts. Nonetheless, AI is beginning to shape broader trends that may gradually influence how sport is practised and experienced.

Table 1. Future trends of AI in sport.

	Trend	Beneficiaries	Types of data	AI methods
AI Use in Elite Sport	Enhanced personalisation	Athletes for customised training and recovery plans, coaches for tailored strategy development	Data from wearables, imaging systems and health sensors	Machine learning, predictive analytics, deep learning
	Digital twin and immersive technologies	Teams for strategy optimisation, athletes for injury prevention and technique refinement	High-fidelity simulation data, sensor data from IoT devices and robotics systems data	Simulation models, computational fluid dynamics, augmented and virtual reality, robotics applications
	Real-time analysis and decision making	Coaches and athletes for instant performance adjustments during events	Biometric data, performance metrics from wearables and sensors	Augmented reality, real-time data processing and feedback systems
	Ethical AI and data governance	Sports organisations for maintaining privacy and compliance, athletes for safeguarding personal data	Personal health records, performance data and biometric information	Encryption, anonymisation techniques and ethical AI frameworks
AI Use in Recreational Sport/Physical Activity	Scalability and automation	Smaller teams with limited resources; fitness enthusiasts	Video footage and repetitive data collection tasks	Automation technologies, video analysis algorithms and machine learning
	Fan engagement and interactive experiences	Fans for enhanced interactive experiences, marketing teams for targeted campaigns	Social media interactions, marketing engagement data	Sentiment analysis, machine learning for personalised marketing strategies
AI Use in Research	Democratisation of talent identification and access	Emerging athletes, sports organisations	Demographic data and regional performance statistics	Data mining and machine learning algorithms for talent identification
	Ethical AI and data governance	Researchers and institutions (compliance and reproducibility)	Health records, performance data and biometric info	Ethical AI frameworks and privacy-preserving techniques
	Digital twin and immersive technologies	Researchers (modelling and simulation of performance/injury scenarios)	Simulation data, sensor data and IoT	Computational models and AR/VR, robotics

One key area of potential lies in enabling more data-driven precision – where the capacity to analyse complex datasets could support more personalised training, tailored health interventions and strategic decision-making informed by evidence rather than intuition.

Another overarching trend is the democratisation of expertise. AI-powered tools make sophisticated analytics and insights accessible to organisations and athletes beyond elite or resource-rich settings. This levels the playing field, fostering inclusivity by identifying and nurturing talent in underserved regions and providing equitable access to advanced training and monitoring systems.

AI also fosters real-time adaptability, a game-changer in both competition and athlete management. Wearable technologies and real-time analysis systems offer instant feedback, enabling on-the-fly adjustments that enhance performance, prevent injuries and optimise strategies during high-stakes events. This dynamic responsiveness transforms how coaches and athletes operate, bridging the gap between preparation and real-world application.

AI promotes sustainability and scalability in sports management. By automating repetitive tasks, such as video analysis or data aggregation, AI reduces time burdens, allowing coaches, medical professionals and administrators to focus on critical decision-making. As these technologies evolve, their integration into sport will only deepen, expanding benefits to recreational athletes, fans and stakeholders at all levels.

To maximise AI's potential, it is vital to address ongoing challenges, such as ethical and privacy concerns, data governance and ensuring equitable adoption across the sporting ecosystem. A careful balance of innovation and responsibility will pave the way for a future where AI enhances human expertise, enabling sport to thrive in a connected, data-rich world.

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